

Dark Patterns, User Behavior, and User Expectations in ChatGPT and Apple Intelligence’s Siri with iOS Permissions

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Abstract

Generative AI assistants such as ChatGPT and Apple Intelligence’s Siri are increasingly embedded into mobile operating systems, where they request broad access to microphones, message history, and other sensitive data through iOS permission flows. Yet it is unclear whether these interfaces genuinely support informed consent or instead exhibit characteristics of dark patterns that nudge users toward sharing more data than they intend. We present an exploratory survey study ($N = 21$) of iOS users that examines how people interpret permissions, documentation, and onboarding screens for ChatGPT and Apple Intelligence’s Siri, as well as their expectations about data collection and control.

Participants completed a four-part online survey featuring realistic screenshots, comprehension and “attention-check” questions, Likert-scale ratings, and open-ended responses. The survey covered: (1) baseline behavior and use of generative AI applications; (2) Apple Intelligence/Siri permissions, training toggles, and related privacy documentation; (3) the ChatGPT iOS onboarding screen and data-use messaging; and (4) demographic information. To understand not only what participants claimed to do but what they actually read, we also deployed a secondary terms-and-conditions (T&C) reading-check instrument embedded in the consent materials. Our analysis combines qualitative coding of free-text answers with descriptive statistics and non-parametric tests over categorical and ordinal responses.

We find that most participants misunderstand which permissions are enabled by default, who can review their transcripts, and how to later disable data-sharing features. For example, no participant correctly identified how to turn off Siri with Apple Intelligence enabled, and 95% reported not knowing how to turn off ChatGPT’s training-related permission. At the same time, participants overwhelmingly preferred opt-in defaults and in-context, plain-language explanations next to individual toggles rather than in separate legal documents: 81% wanted Apple’s documentation to appear directly alongside the relevant permission, and 19 out of 21 participants preferred opt-in for Siri-related permissions that send data off-device or train

AI models. Although participants could clearly distinguish when location access is functionally needed (e.g., for weather) versus unnecessary (e.g., for a simple math query), current systems often treat these cases identically, suggesting a misalignment between user expectations and technical reality.

Taken together, our findings suggest that existing permission flows for mobile generative AI assistants exhibit interface interference and semantic ambiguity that weaken informed consent. We discuss how small design changes—such as inline “key facts” summaries, clearer distinctions between functional and training access, more intuitive placement of controls, and visible indicators of human review—could reduce dark-pattern-like effects and better align AI permission flows with users’ privacy expectations.

1 Introduction

Permission prompts play a central role in how users make privacy and security decisions on digital platforms. In an ideal “notice and consent” model, these prompts help users understand which data is being requested, for what purpose, and under what conditions it may be retained or shared, so that they can make informed choices about whether to grant or deny access. In practice, however, permission flows frequently fail to live up to this ideal. Prior work on dark patterns and mobile permissions has shown that vague wording, visually dominant confirmation buttons, default opt-ins, and non-intuitive settings can manipulate users into granting broader permissions than they intend, even when users care about privacy and would prefer more restrictive defaults [8, 17, 18, 34].

The rapid deployment of generative AI assistants has intensified these concerns. Apple Intelligence brings large-scale AI capabilities directly into iOS and powers features such as Apple Intelligence’s Siri, while stand-alone mobile apps such as ChatGPT offer conversational assistance on the go. To provide these services, AI assistants often request extensive permissions, including access to microphones, transcripts, app activity, and sometimes location information. At the same time, the underlying models may store, review, and reuse con-

versational data to improve system performance. For ordinary users, a single tap on a seemingly harmless “Continue” or “Improve” button can authorize long-term retention and cross-context use of intimate conversations, with little visibility into what actually happens behind the scenes.

Research on smartphone permissions has repeatedly found that users rarely read privacy policies, struggle to form accurate mental models of data practices, and are highly sensitive to interface cues when deciding whether to share information [13, 14, 38]. Users may glance at a dialog, infer its meaning from a few keywords and layout choices, and press the most visually salient button. Work on dark patterns further shows that designers can exploit these tendencies through “interface interference,” where the interface visually highlights the easy path over the informed path, and through “obstruction,” where important controls are hidden deep in settings menus [17, 18]. Yet most existing studies focus on Android or on general app permissions, rather than on generative AI assistants embedded in iOS and the particular consent challenges they introduce.

At the same time, recent work on privacy documentation and app privacy labels suggests that simply adding more text is not enough [35, 43]. Users are overwhelmed by technical details, confused by ambiguous terminology, and unwilling to invest the time needed to read long policies. Terms such as “subset of interactions” or “may be used to improve our services” appear transparent but often leave users unsure about who exactly reviews their data, how long it is stored, or how to opt out. In the context of generative AI, where transcripts and audio may capture highly sensitive information, this ambiguity can significantly undermine user trust.

Against this backdrop, it is not sufficient to ask whether users technically “consent” to data collection. Instead, we must examine whether current AI permission flows actually communicate what is happening to users’ data, whether users understand how to exercise control, and how design choices may be nudging them. Mobile implementations of ChatGPT and Apple Intelligence’s Siri provide a timely and concrete case study: they are widely recognized brands, they operate within the same iOS ecosystem, and they rely on similar combinations of permissions, privacy notices, and documentation.

In this paper, we investigate how users interpret and engage with the permission and documentation flows of ChatGPT and Apple Intelligence’s Siri on iOS. We focus on three overarching research questions:

- **RQ1:** How do users understand the permissions, data flows, and review practices associated with ChatGPT and Apple Intelligence’s Siri on iOS?
- **RQ2:** To what extent do current permission and onboarding interfaces support informed consent versus exhibiting dark-pattern-like characteristics such as interface interference, semantic ambiguity, or obstructed controls?

- **RQ3:** What types of interface changes and default settings do users prefer to improve transparency, control, and comfort when using mobile generative AI assistants?

To explore these questions, we conducted a mixed-methods online survey with 21 iOS users. The primary survey instrument contained four sections: (1) consent and baseline behavior, (2) an Apple Intelligence/Siri module, (3) a ChatGPT module based on the iOS onboarding screen, and (4) demographics. Participants answered multiple-choice questions, Likert-scale items, and short free-text prompts about their expectations of default permissions, their interpretations of specific permission language, and their beliefs about who can review their data. We also asked targeted questions to probe users’ mental models of data necessity (for example, whether Siri needs location to answer “What is the weather today?” versus “What is 1+1?”), as well as their preferences for opt-in versus opt-out defaults and for in-context versus separate documentation.

In addition, we deployed a secondary T&C reading-check instrument embedded within the participation terms themselves. This allowed us to compare participants’ self-reported behavior (e.g., “I read the terms”) with their actual attention to consent text, and to understand how much traditional “notice and consent” mechanisms are functioning in practice. Our analysis combines qualitative coding of open-ended responses with descriptive statistics and non-parametric tests (Fisher’s Exact Test, Mann–Whitney U, and Kruskal–Wallis) over categorical and ordinal data.

Our findings paint a nuanced picture. On one hand, participants generally recognized that AI systems can train on user data and that not all tasks require sensitive permissions; for instance, all participants correctly responded that Siri does not need location access to compute a simple math query. On the other hand, participants were often unsure which permissions are automatically enabled, who can review their conversations, or how to later disable AI-related features. No participant selected the correct method for turning off Siri with Apple Intelligence enabled, and almost all participants reported that they did not know how to turn off ChatGPT’s training-related permission. Most participants failed the attention checks embedded in the T&C document, yet they strongly expressed a desire for clearer, in-context explanations: 81% wanted Apple’s documentation to appear directly alongside individual permissions rather than in separate pages, and 19 out of 21 participants preferred opt-in defaults for Siri-related permissions.

These results suggest that current permission flows for generative AI assistants behave in ways that resemble dark patterns: they foreground a large “Continue” button, rely on opt-out or opaque defaults, use ambiguous terminology, and bury critical information in dense documentation that users rarely read. At the same time, our data show that users are not indifferent to privacy; they care about who can see their transcripts and want straightforward controls that match their

expectations.

In summary, this work makes three contributions. First, we empirically characterize user understanding, expectations, and misconceptions regarding iOS permission and onboarding flows for ChatGPT and Apple Intelligence’s Siri. Second, we identify specific interface features that exhibit dark-pattern-like properties and show how they affect user behavior and perceived control. Third, we propose design implications for more trustworthy AI permission interfaces on mobile devices, including clearer separation between functional and training access, inline “key facts” summaries, and more discoverable opt-in controls.

2 Background and Related Work

2.1 Smartphone Users’ Access-Control Needs

The literature on smartphone permissions, users’ interaction with these permissions, and access-control policies put in place are extensive; Unfortunately, most of these studies are done on the Android platform due to it being open source and easier to access in terms of restrictions. However, these studies can still be helpful in giving insight to user behavior and expectations. Research done on Android app access-control and permission reviews and user experience has found users typically don’t pay attention to permissions when asked and if they do, they have a low understanding of the permission and it’s scope [13, 37]. There have been behavioral studies done when a user interacts with a permission. These studies revealed that the context in which a user interacts with a permission, the user’s perception of the sensitivity of the permission, and the user’s expectation of the permission all influences a user’s choice to accept or deny the permission [6, 8, 34]. Shih et al. [38] revealed that if a permission gives no explanation or context for the purpose of data use or data access, users were actually more willing to disclose data. There have also been studies that revealed many users aren’t even aware of certain privacy and security settings or their defaults, revealing that many users haven’t configured any of these types of settings in the past due to difficulties in usability [14]. In the IOS space, research has found that users are concerned with permissions that violate their trust, permissions that allow access to sensitive data, and users’ confusion about their understanding of certain Apple alerts [30, 42]. There has been research that documented the users’ discomfort with the “creepy” ways in which apps interacted with their data [40]. These concerns may not be completely unprompted, Egele et al. [12] conducted a study where they developed a tool that analyzed the data flows of over 1,400 iPhone applications, where they found that over half of the applications leaked the device’s unique ID. Additionally, Agarwal et al. [2] found that access to sensitive permissions, such as location and the user’s unique identifier is common within iOS applications. There have been numerous studies that documented that there

are instances where applications request permissions that are not integral to the app’s functionality [2, 10, 26, 38].

2.2 Privacy Policies and Documentation

Wu et al. [43] conducted interviews with Apple users regarding “Apple’s App Privacy Report” documentation, noting that users knew how to remove access to an app, however users were still overwhelmed and ended up confused by the technical details within the documentation. Typically, data access information is conveyed within the Privacy Policy of an application. However, many studies have found that these Privacy Policies aren’t read by users, and if they are, they’re hard to follow along by users [9, 16, 35]. In 2008, McDonald et al. [29] noted that it would take about 244 hours per year for a user to read all the privacy policies for all the websites that the user visits. And, this number has only increased as there’s been a progression towards the Internet of Things(IOTs) and as the internet and the number of websites a user will visit expands [35]. There’s been research done to automate this process of reading privacy policies, however this automation has not been perfected as of yet and still displays some misclassifications [21]. Recently, there has been a push for app privacy labels that try to summarize these data practices in a short, succinct way. There has also been multiple studies done on these privacy labels that show users still being unsatisfied and misinformed about these privacy labels [1, 39].

2.3 Dark Patterns In Permissions

There has been extensive research on dark patterns in permissions. These studies have analyzed and categorized dark patterns in broad categories of user interface designs [7, 17, 18, 23]. Prior work continued on dark patterns particularly used by mobile app developers that may mislead users into taking unwanted actions [15, 19, 30].

2.4 Generative AI Application Mobile Device User’s Permissions and Access-Control Needs

In this paper, we focus on dark patterns used by ChatGPT and Apple Intelligence’s Siri that fall under our categorizations of dark patterns and their effects on user behavior and expectations. The extreme rise in popularity of AI and in particular, chat bots, also brings the rise of concerns of users’ privacy and security when using AI. Previous works have studied the security implications of using AI chat bots [20, 24, 36] and user perception of security and privacy of AI [41], while one study found that only about 4.35% of chat bots that used permissions provided a privacy policy [44]. There is little work on AI permissions and dark patterns or how they’re perceived and used by users, especially when it pertains to AI chat bots. Malkin et al. [28] conducted an interactive simulation that

simulated an experience of a proactive assistant, an AI agent that continuously records conversations to offer suggestions to users without being directly invoked, between a pair of participants having a conversation, and had the assistant display a permission prompt related to data the assistant needed access to offer a suggestion. The study showed that users could be described as receptive or enthusiastic about the use of a proactive assistant, even when they were aware of the privacy concerns. They also found users typically would prefer an interruption-free session rather than being able to control the permissions the assistant was able to obtain. However, there are available blogs that speak about AI and chat bot permissions [25, 31], with some even focusing on iOS permissions while using these AI application [22]. However, these blogs do not research nor study any relationship between users and AI permissions, they are simply sources of information. These blogs are also not official or academic resources, so their reliability cannot be confirmed and may be limited in this regard. Our study hopes to close the gap between AI chat bot applications and permissions on mobile systems, specifically iOS.

3 Methodology

3.1 Study Design and Overview

We conducted a mixed-methods, online survey study to examine whether iOS permission and onboarding interfaces for mobile generative-AI assistants support informed consent or instead exhibit characteristics associated with dark-pattern-like design (e.g., interface interference, semantic ambiguity, and obstructed controls). The study focused on two iOS contexts: (1) Apple Intelligence/Siri privacy and permission controls, and (2) the ChatGPT iOS onboarding screen and associated data-use messaging. All interactions were survey-based: participants viewed static screenshots and answered questions about intended actions and interpretations rather than being observed in-app. To measure both (a) participants' interpretations and preferences and (b) whether participants read consent materials, we used two instruments: (i) a primary survey containing scenario questions and screenshots, and (ii) a secondary terms-and-conditions (T&C) reading-check followed by a brief debrief survey.

3.2 Participants and Recruitment

Participants were adult volunteers recruited to complete a Google Forms survey. The analyzed dataset includes $N = 21$ respondents. We collected non-identifying demographics (e.g., age range and education level) and self-reported technical background for descriptive summaries and exploratory subgroup analyses.

Operational definitions for subgroup analysis. We defined subgroup thresholds *a priori* based on survey Likert-scale items. *Technical background* was defined as a score ≥ 3 on a 5-point scale assessing how technical a participant's education/profession is. *Familiarity* with an application (Siri with Apple Intelligence, and ChatGPT) was defined as a score ≥ 3 on a 5-point familiarity scale (binary: familiar vs. not familiar).

3.3 Ethics, Consent, and Debriefing

Participants reviewed a participation terms document describing voluntariness, time estimate, anonymity, and data handling. To reduce priming effects (e.g., prompting participants to scrutinize interfaces specifically for manipulation), we used authorized deception by presenting a general study purpose initially. After participation, we provided a debriefing statement explaining the true purpose of the study and included an option for participants to withdraw their responses post-debrief.

3.4 Materials and Measures

3.4.1 Primary Survey Instrument

The primary survey included four components: (A) consent/eligibility, (B) an Apple Intelligence/Siri module, (C) a ChatGPT module, and (D) demographics.

Apple Intelligence/Siri module. This module measured (i) participants' expectations of default-enabled permissions (multi-select), (ii) interpretations of permission language and documentation excerpts (free response), (iii) mental models of contextual data necessity using paired tasks (e.g., "What is the weather today?" vs. "What is 1+1?" for location necessity), (iv) preferences for permission defaults (opt-in vs. opt-out), and (v) procedural knowledge of how to disable Siri with Apple Intelligence.

ChatGPT module. Participants were shown a screenshot of the ChatGPT iOS onboarding/welcome screen (Figure 1) and answered (i) intended next action (e.g., press "Continue" vs. open "Learn more"/policy links), (ii) what the screen communicates (multi-select comprehension), (iii) expectations about training and review (categorical items), (iv) where they would expect to learn more about data use, (v) Likert ratings of perceived clarity/informedness/caution, and (vi) which UI elements influenced their decision-making (checkbox list including the large "Continue" button, headings, iconography, and link placement).

3.4.2 Secondary Instrument: T&C Reading-Check and Debrief Survey

To test whether participants read consent materials rather than merely self-reporting, the participation terms included embedded instruction-based attention checks (e.g., specifying an exact phrase to enter when later asked a particular question). After the primary survey, participants completed a brief debrief survey that asked whether they followed those attention-check instructions and, if not, why (e.g., did not read, skimmed, forgot).

3.5 Procedure

Participants completed the study remotely in three steps: (1) reviewed the participation terms and consented, (2) completed the primary survey modules (Apple Intelligence/Siri and ChatGPT), and (3) completed the debrief survey and reviewed the debriefing statement, with an explicit post-debrief option to withdraw their data from analysis.

3.6 Data Analysis

We analyzed survey responses using a mixed-methods approach combining descriptive quantitative summaries, an exploratory subgroup analysis, and qualitative coding.

Quantitative summaries. For each closed-ended item, we report counts and percentages. For multi-select questions (e.g., intended next action and which UI design elements influenced decisions), we summarized each option as the number (and proportion) of participants selecting that option. For 5-point Likert items (e.g., perceived clarity/informedness/caution), we treated responses as ordinal and report distributional summaries (frequency by scale point) rather than assuming interval properties. Participants could skip items; therefore, denominators vary by question and missing responses were excluded pairwise.

Familiarity-based subgroup comparisons. To examine whether prior familiarity with an application relates to participants' behaviors and expectations, we operationalized familiarity as a binary indicator based on the 5-point familiarity item (≥ 3 vs. < 3). For categorical outcomes (including binary indicators derived from multi-select items and coded qualitative outcomes), we tested for association between familiarity group and response using Fisher's Exact Test due to small cell counts. We used a significance threshold of $\alpha = 0.10$ to increase sensitivity in this small-sample exploratory study and report exact p -values for each comparison. We conducted subgroup tests only when both familiarity groups had sufficient sample size.

Qualitative coding (codebook). Short-answer and paragraph-style responses (e.g., interpretations of permission language, expectations about who/what can review data, and rationales for opt-in/opt-out preferences) were analyzed via a codebook. We iteratively developed mutually exclusive categories for each open-ended question based on recurring themes, then applied the finalized codes to all responses. We report category frequencies and use representative themes to contextualize the quantitative patterns.

Methodological note. Because the study used survey questions and static screenshots to simulate user interactions, results should be interpreted as measuring comprehension and intended behavior rather than fully capturing real-time in-app behavior.

4 Results

We present our findings based on the two topics we are trying to analyze, user behavior, user expectations, and possible relationships, each divided into the two applications we are analyzing, Apple Intelligence's Siri and ChatGPT.

4.1 Data Summary

We collected data from 21 participants. 13 Participants were men, 7 were women, and 1 was non-binary. 20 participants were in the 18-25 age range with 1 participant being in the 26-35 age range. 3 participants were White, 17 were Asian, 1 was Black, 1 was Hispanic or Latino, and 1 was native Hawaiian or other pacific islander. 1 participant noted that they are a part of the LGBTQIA+ community. 20 participants have achieved or in the process of achieving higher education (Bachelors, Master, Ph.D.) with the other participant having achieved high school education. We used a 5-level Likert scale to ask how technical a participants profession or education is (1: not technical at all, 5: extremely technical). We will define a participant who is in a technical profession or education as a participant who answered 3 or higher on this scale. Using this definition, 16 participants identified themselves as working in a technical environment, while 3 didn't, and 2 others didn't choose to provide this information. 10 participants noted that their income is less than 75,000\$, 2 said their income was above 100,000\$, 7 said they didn't have an income, while 2 chose not to provide this information. We used a similar 5-level Likert scale as the one described earlier to identify users who are familiar with Apple Intelligence's Siri and ChatGPT (1: I have never used the application, 5: I use the application very often). We will also use the same definition as before where we will identify a participant as familiar with the application if they answered 3 or higher on the scale of familiarity. 10 users noted being familiar with Apple Intelligence's Siri and 20 users noted being familiar with

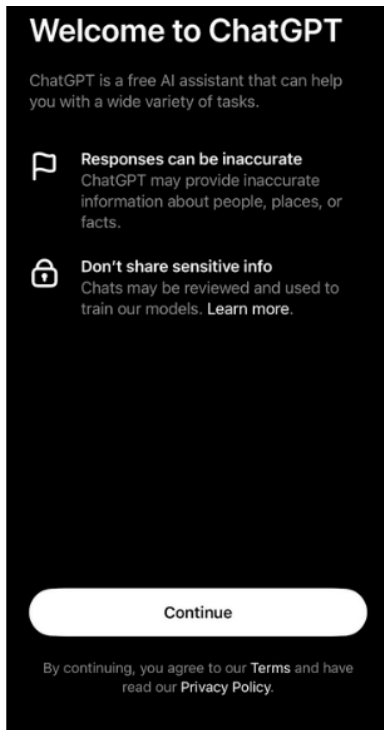


Figure 1: ChatGPT’s mobile application’s welcome page

ChatGPT, with 17 out of all the participants have downloaded ChatGPT’s mobile app.

4.2 User Behavior

4.2.1 Apple Intelligence’s Siri

We asked the participants if there were any or would have been any permissions they specifically turned on or off after starting to use Apple Intelligence’s Siri. Only 1 of the participants noted changing any permissions and still using Siri after. This 1 participant documented turning off any location or tracking permissions, while 2 other participants stated they turned Siri off all together, but stated nothing about any other permissions.

4.2.2 ChatGPT

We showed participants Figure 1, which is a screen cap of the welcome page and first thing you see once downloading the ChatGPT mobile application on iOS. We then asked if shown this screen, what the participants would do next in a multi-select fashion with the options:

- Press Continue
- Close the app
- Click "Learn more"

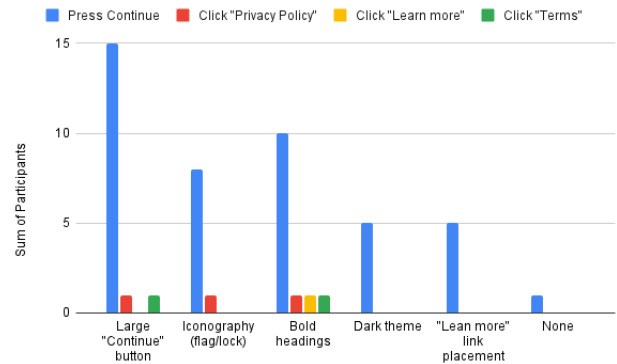


Figure 2: Design Influences of Participants Based on Their Behavior

- Click "Terms"
- Click "Privacy Policy"
- Other:

18 participants said they would just "Press Continue", while 1 participant said they would click "Learn More". 1 participant said they would press "Terms", "Privacy Policy", and "Press Continue", while 1 other participant said they would press "Privacy Policy" and "Press Continue." We also asked participants what part of the design impacted their decision-making. For all the participants who chose "Press Continue," the large "Continue" button was the most prevalent influence as 15 out of the 20 chose this option, with bold headings being noted by 10 users, followed by iconography (flag/lock) being noted by 8 participants. Other options like the dark theme and the "Learn More" link placement were the least prevalent with both only being noted by only 5 participants each. For the participants who said they would have different behaviors besides pressing continue, the large continue button and the bold headings still seemed to be the most popular design influences.

4.2.3 Survey Terms and Conditions

Our team created a Terms and Conditions/Privacy Policy for the survey itself. Within this document, we added 3 attention checks. These attention checks told users if they see a specific question, they should answer in a specific manner. The 3 attention checks were spread out within the document, with 1 at the beginning, 1 in the middle, and 1 at the end. The 3 attention checks were displayed as follows within the documentation, in order:

- If you read this then when you see the question “What other generative AI applications do you use?”, answer with “To whom does this pertain to.”

- If you see the question “How do you spend your free time” on the survey, respond with “watching paint dry”.
- If you see the question “What’s better — ChatGPT or other AI applications?” on the survey, answer with “ChatGPT will never be beaten.”

17 people didn’t put the correct results for any of the Terms and Conditions attention checks. After the survey, we linked another survey for the participants to be debriefed about the Terms and Conditions. 14 people participated to this "debrief survey." 8 participants said they didn’t read the Terms and Conditions while 3 users said they skimmed over it so they didn’t catch any details. 1 user said they read the informed consent document thinking it was terms and conditions. The final 3 users read the terms and conditions and proved it with providing the correct answers for all the attention checks. Based on the participant’s answers to the attention checks, none of the participants only got some of the attention checks and didn’t get some others, so if participants knew 1 attention check, it’s guaranteed that they’ll know the rest. These results may mean that people who choose to read documentation at all tend to read all of it, instead of just some of it. However, our Terms and Conditions are really short(1 page), if it was longer this assumption may not be true, as is the case with most documentation for applications. Overall, these results point to the fact that users tend not to read the documentation at all or they skim over it, but still miss a lot of details, when agreeing to use an application, which is a big problem, if applications decide to put most, if not all, important information for users inside the documentation.

4.3 User Expectations

4.3.1 Apple Intelligence’s Siri

We first wanted to grasp user’s current expectations about the default settings of iOS permissions when using Siri, so we gave a multi-select question and asked participants which permissions do they think are automatically turned *on*. Here are the options we gave to the participants:

- Siri may train itself based on your data
- Transcripts and Audio Transcripts between you and Siri are allowed to be stored on Apple Servers
- Location Access
- Siri may listen passively at all times in case you may need them
- All transcripts and audio transcripts must stay locally on your device
- Siri may look at what apps you’re using to learn more about you

- Transcripts of all search requests you ask Siri(i.e. asking Siri to look something up on Safari)
- Other:

The reality of the permissions are as follows:

Opt-in

- Siri may train itself based on your data
- Transcripts and Audio Transcripts between you and Siri are allowed to be stored on Apple Servers

Opt-out

- Location Access
- Siri may look at what apps you’re using to learn more about you
- Transcripts of all search requests you ask Siri(i.e. asking Siri to look something up on Safari)

Not real permissions

- Siri may listen passively at all times in case you may need them
- All transcripts and audio transcripts must stay locally on your device

The most prevalent option amongst the participants was "Siri may train itself based on your data," although this permission is actually opt-in (off by default). This suggests users are more aware of this type of permission, perhaps due to generative AI applications. A handful of users also said "Transcripts and Audio Transcripts between you and Siri are allowed to be stored on Apple Servers" is on by default; however, this permission is tied to the Siri training permission, so it is also off by default. Despite these two permissions being linked, only 7 users said both, suggesting a gap in understanding that giving permission for Siri to train on data also means user transcripts are stored on Apple servers. Most people chose at least one of the three real opt-out permissions, showing some level of understanding of which permissions are opt-out. However, no users were able to list all three opt-out permissions, either listing a subset or mixing them with incorrect options, which may show a significant gap in understanding and expectations of opt-out permissions. The option "Siri may listen passively at all times in case you may need them" is not a real permission users can control or Apple claims Siri can do. However, this permission was chosen by almost half of the participants, which may show a gap in expectations of what Siri has access to while turned on. The other "not real" permission, "All transcripts and audio transcripts must stay locally on your device," is also not controllable by users. Due to Apple’s Private Cloud Compute capabilities, requests that

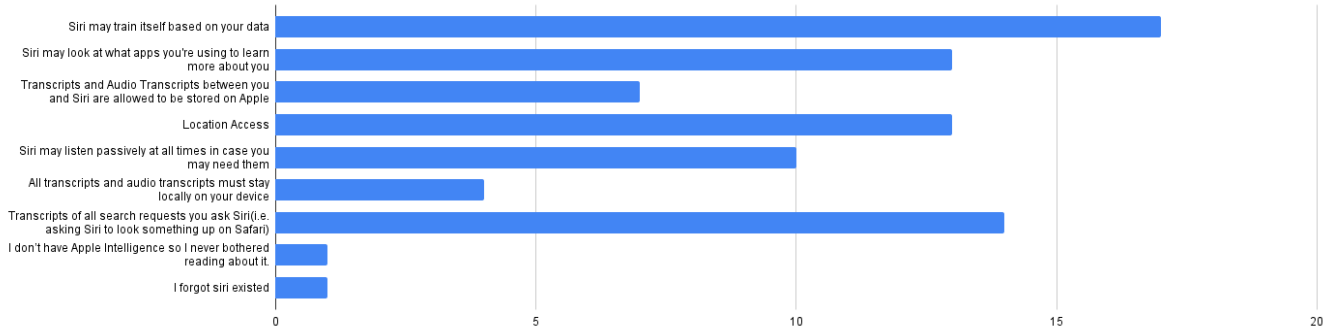


Figure 3: Which Permissions Participants Expect Are Defaulted On

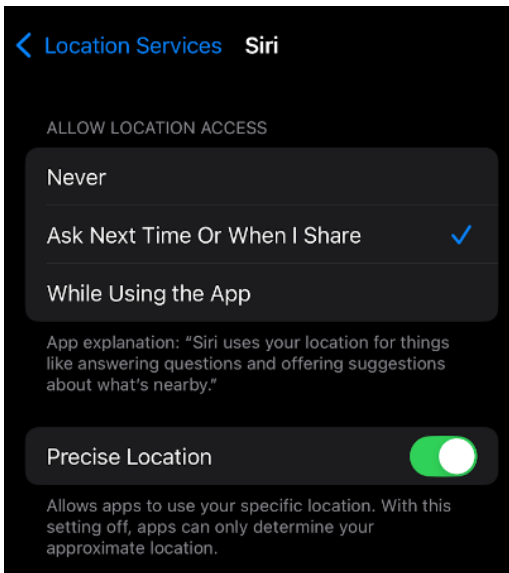


Figure 4: Apple Intelligence's Siri's Location Permission

cannot be handled on the device are sent out to Apple's Private Cloud. Only 4 participants chose this option. This could be because most users don't understand that data leaving your device is a possibility, or perhaps because they understood the option but didn't think it was an opt-out permission.

We showed Figure 4 to the participants and we asked them what they think it means to have the permission on. We allowed users to give a short answer description, so to analyze the qualitative results, we created a code book to organize the answers by a code, as shown in Table 2. Using the code book, we noted that around 57.14% of users mentioned that the permission must be asked or must be given consent to before Siri can use the user's location. However, even though many users mentioned this, the location permission is an opt-out permission, therefore it is never asked to be turned on when the user starts using Siri. This may show a gap between the user's expectations on which and when permissions are given permission to use the user's data and

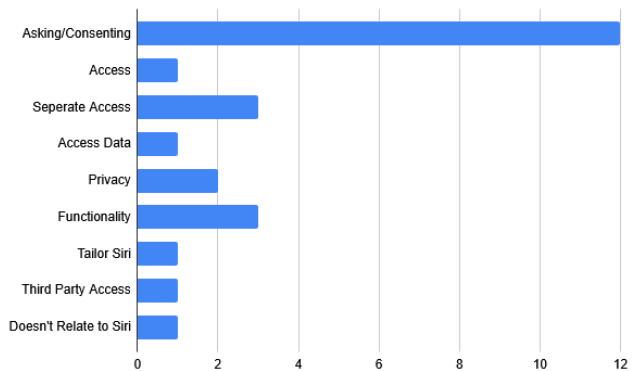


Figure 5: Coding of Users Expectations with Siri's Location Permission

what techniques Apple currently employs. 3 participants also mentioned functionality in some way, mentioning that the permission will allow Siri to access a user's location when a request needs the user's location in order to be completed, giving a sense that some users believe that their data is only accessed when needed for functionality, however as found in previous works, this may not always be true [2, 10, 26, 38].

Data accesses that aren't integral to the application's functionality may actually also be an issue that Apple Intelligence's Siri is guilty of as well. Yoav Madrid, a security researcher, at Lumia Security recently wrote an article and presented it at Defcon 2025 called "AppleStorm." In his research, he investigated the network traffic from Apple Intelligence's Siri after asking it a request. He tested different queries that couldn't be handled locally on the device and had to be sent outside the device, and he found that if the user has the location permission enabled, then the user's location is appended to every request the user makes, regardless if the user's location is needed for functionality [27]. So, in our research we wanted to test user's expectations when they have the location permission enabled and when their location

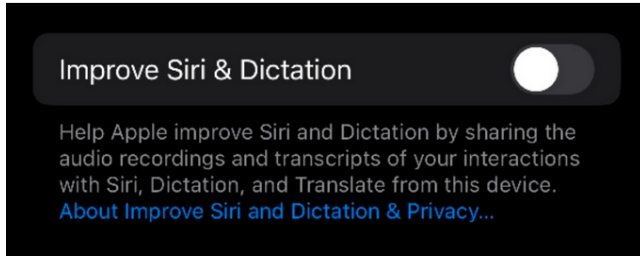


Figure 6: Apple Intelligence's Siri's Improve Siri and Dictation Permission

is accessed by Siri. We asked the participants if they think their location needs to be accessed for the request "What is the weather today?" and "What is 1+1?" For the former question, 100% of the participants said "Yes." However, for the latter, 100% of the participants said "No." So, if it is indeed true that Siri accesses the user's location for *every* request, this may be a big privacy concern for users or is at least a feature that users are not expecting at all, that if they did know this information, they may be more cautious about turning on the location permission.

We showed participants Figure 6, which is the permission that allows Siri to train itself using the user's data. We asked participants what they think it means to have this permission on to gauge user's understanding and expectations. This question allowed participants to give a short paragraph qualitative answer, so we created a code book for the participants answers in order to organize them in Table 3. We have three categories within our code book (Improvement, Broad Training or Learning, Individual Training or Learning) that if the participants' answers are under one of these three categories, we will take it as they understand the overall message of this permission, that Siri can improve or train itself on the user's data. Almost all the participants understood the main concept with 18 participants' answers being under one of the three codes above. However, when this permission is turned on, Siri can train itself to improve itself as a whole and to personalize the user's experience [5]. Most participants understood that Siri trains itself to improve its product as a whole, but only 4 participants said Siri can use your data to personalize the user's experience, and 0 participants said that Siri can both train itself to improve the product as a whole and to improve the user's personal experience. This may show that there is a gap in understanding for many users of the full scope of what Siri does when it trains itself on the user's data. There were also 2 participants who's answers were under the "improvement" code, but also the "listening" code, which means that they thought this permission meant that Siri can passively listen to the user at all times or some of the time to then train itself on what it hears, which is not something

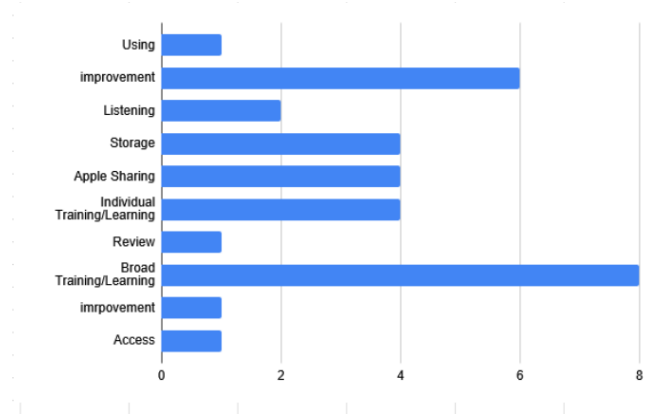


Figure 7: Coding of Users Expectations with Siri's Improve Siri Permission

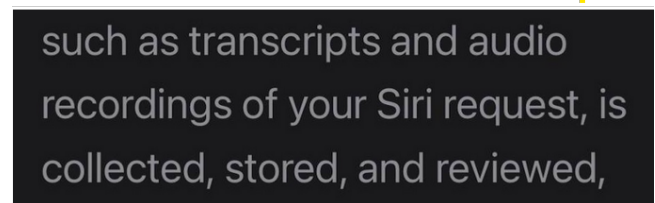


Figure 8: Apple Intelligence's Siri's Privacy Policy About Data Use

Apple says it does, but this shows a trend from the previous question about what permissions participants thought were on by default and how some participants chose that "Siri may listen passively at all times in case you may need them." This trend where participants think that Siri can listen passively while not being in active use, may show a larger issue where many users think Siri has the ability to do this and Apple may be misinforming or not informing their users enough on what their applications can actually do. We also wanted to study if users understand the underlying conditions with this training permission, that turning on this permission means that Apple can store your audio recording and transcripts and can be reviewed by other entities. 4 participants' answers were under the code "Apple Sharing," which means they understand their data is shared with Apple and not kept locally on their device, however these participants did not note that their data can be *stored*, which may be a bigger privacy concern than simply sharing their data, which doesn't imply that their data can be saved for later use. So, only 4 participants mentioned that this permission means that Apple can store their data on their servers, while only 1 participant mentioned the act of their data being reviewed, which may show a lack of understanding and incorrect expectations when users turn on this permission, where if they knew this information then their decision to turn on this permission may change.

recognition models. Apple may also review a subset of the transcripts of your interactions with Siri, and this small subset of interactions that has been reviewed may be kept beyond two years for the ongoing improvement of these products and services.

Figure 9: Apple Intelligence’s Siri’s Privacy Policy About Data Use for a Subset of User’s Data

We also wanted to study the participants understanding and expectations of Apple Intelligence’s privacy policy. We showed participants Figure 8 and Figure 9 which is from the privacy policy linked in the Apple Intelligence & Siri settings called "About Siri & Privacy". We allowed the participants to give a short paragraph qualitative answer, in which we created a code book out of in Table 4. Within Apple’s "Improve Siri and Dictation & Privacy" privacy policy, which is a separate privacy policy than before, it states that "Only Apple employees, subject to strict confidentiality obligations, are able to access audio interactions with Siri and Dictation" [5]. 9 participants understood that their data is reviewed by Apple or by Apple employees in a broad sense, but they did not give specific information about who exactly within Apple can look at their data, indicating that the participants don’t know who specifically within Apple or they think any kind of person working under Apple may look at their data. 5 participants didn’t mention an Apple employee at all, 2 users said it was reviewed by a third party, while the other 3 said it is reviewed by some kind of technology i.e. an algorithm or Apple systems. 5 participants did mention only special type of Apple employee being able to review the data, specifically ones who work in Apple Intelligence or Apple Intelligence’s Siri, but none of them mentioned the confidentiality obligation aspect. One of the participants did say "trained Apple employees," which could mean the Apple employee is trained to keep the data confidential, but the answer isn’t specific enough to confirm that assumption. So, this could indicate that if the user only read the broad "Improve Siri and Dictation & Privacy" privacy policy instead of the specific, separate "Improve Siri and Dictation & Privacy" privacy policy, they would misunderstand who handles and can review their data. This kind of practice where privacy policies are separated into many different documents and put in many different places within the settings of the application, could indicate a dark pattern of obstruction where it is difficult for users to find specific information

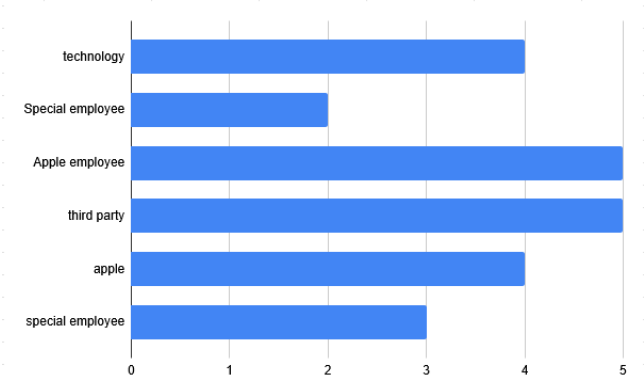


Figure 10: Coding of Users Answers For Who They Think Their Data Is Reviewed By

because of its difficulty to access the information, causing users to end up misinformed or confused.

For Figure 9, we asked participants if they could define what a subset it and how it’s chosen based on what they read. Our research team couldn’t find out the specific details of what a subset is and how they’re chosen based on the privacy policies alone, so we wanted to know what the participants thought or expected when reading this information. The participants were allowed to answer with a short paragraph qualitative answer coded with the code book in Table 5. 5 participants said that the documentation doesn’t specify this kind of information or that they don’t know the answer. Some did, however, give answers on how a subset is chosen, with 1 participant saying a subset is chosen based on how something is being said like the tone of the voice or what language is being spoken and 4 participants mentioned it could be chosen by if Siri is not able to fulfill a certain request correctly or Apple wants to train Siri on a specific kind of topic/question, a subset pertaining to that area will be kept to improve Siri in that area. What’s interesting, however, is that some participants did note that Apple could just end up using all of a person’s data, because there is no specification on how large a subset could be, with 1 participant saying "...they’re just going to use all of it." 1 more participant noted "My data is being stored no matter how long and how would I know it’ll be just 2 years?," which the documentation also didn’t specify by just saying "...may be kept beyond two years..." So, we have 5 participants who aren’t sure how to deduce the documentation and we have another 5 participants who are questioning Apple’s sincerity and who’ve deduced that because of the documentation’s vagueness, that Apple could technically overstep privacy boundaries. This could elude to Apple’s privacy policies being innately obscure and cannot be understood correctly by many users.

To understand if users are complicit with this, and not

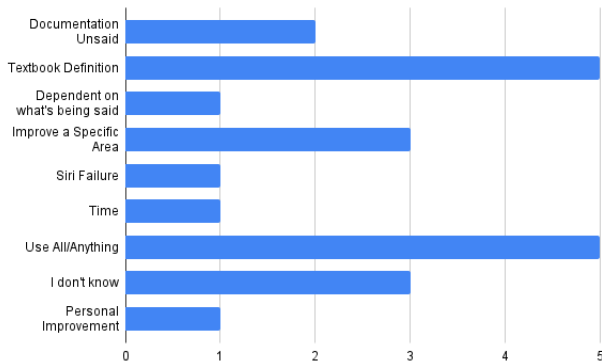


Figure 11: Coding of Users Answers For How They Think A Subset Is Chosen

just how they understand Apple's documentation, we asked participants if they wish for the information in Apple's documentation to be showed with the permission itself(i.e. in the short caption under it). Or, is it within Apple's right to put it in a separate document for users to choose whether to read it or not. 81% said they wished for this information to be shown with permission itself. So, regardless if the user is correctly understanding the permission and even if they read the privacy policy, an overwhelming number of users wish this information to be more easily accessible by having it listed with the permission, showing that many users do prioritize accessibility and wish for a more accessible reality.

Finally, we wanted to know if user's have specific preferences/expectations for permission defaults with Apple Intelligence's Siri. We asked participants if they think that for any permissions that may pertain to Apple Intelligence's Siri, do they think they should be Opt-in or Opt-out? And, why? Participants were able to give a short paragraph qualitative answer so we created a code book, shown in Table 6. Many people didn't give a reason why, but 19 participants said they wished for permissions to be opt-in, with none of the participants only giving specific permissions, so our research team is assuming that these participants wish for **all** permissions to be opt-in. For the users who wished for permissions to be opt-in, the most popular reason is that participants wish they are knowledgeable about the applications they are using and the data they access, with one participant specifying it is their right, saying "I have right to determine which one should be turned on rather than turned on by default." 3 participants also mentioned privacy being important to them, and one participant saying their reason is because these permissions mean the user would have to share sensitive data, showing that participants see permissions being opt-out as a possible breach of privacy. Any participant who said opt-out did not give a reason as to why.

We also asked participants if they think permissions relating to Apple being able to send your data outside of your device [3] and permissions for Siri to have access to information that Siri doesn't need to function but may make Siri "work better" should be opt-in or opt-out. 81% said "opt-in" for the former. Currently, Apple doesn't give an option for users to allow their device to send data outside of the device or not, but the participants desire for this to be an opt-in permission along with some participants believing that this was a permission in the first place in the previous survey question "What permissions do you think or that you are aware of that Siri has that are automatically turned ON after downloading Apple Intelligence ?," can be interpreted to mean that users wish for this type of granularity in their permissions and wish to be informed about these types of actions that their device and application can take. 85.8% said "opt-in" for the latter. For the latter, 1 participant filled out the "other" option, saying "Like better for the certain user, yes." We found this to be an inconclusive option, as we're not sure whether the participant was saying yes to opt-in or opt-out, however this specification could allude to that some users think the default nature of a generative AI application training itself on data permission depends on *why* the application is training itself on the data. Overall, these results shows that users do wish for permissions to default to off, instead of on, due to concerns with privacy, lack of knowing, and concerns with access to sensitive data.

Finally, we asked participants if they even knew how to turn off these Apple Intelligence's Siri. We gave this as a multiple choice question with these options:

- Turn it off in Privacy Security Settings
- Turn it off in Siri Settings
- Turn it off in Content Privacy Restrictions settings in Screen Time settings
- I don't know, but I would search online
- Other:

The correct answer is "Turn it off in Content & Privacy Restrictions settings in Screen Time settings" [4]. However 0% of participants selected this option. 61.0% of users said they didn't know, and the remaining 38.1% of participants gave the incorrect answer. This shows a big gap between the actuality of where permissions are placed and where user's expect this permission to be. This could be an example of how Apple obfuscates the placement of permissions by placing them in non-intuitive placements to make it more difficult for users to turn off the application or for users to control the application they're using in the way they want.

4.3.2 ChatGPT

After showing participants Figure 1, we asked the participants what this screen is telling them with a multi-select question with the following options:

- ChatGPT may be inaccurate
- Don't share sensitive info
- Chats may be reviewed by humans
- Chats may be used to train models
- It needs microphone/camera access
- None of the above / Not Sure
- Other:

The correct grouping of these options are:

The screen explicitly says:

- ChatGPT may be inaccurate
- Don't share sensitive info

The screen doesn't explicitly say it, but it's information pulled from ChatGPT's documentation:

- Chats may be reviewed by humans [33]
- Chats may be used to train models [32]

Not a real permission:

- It needs microphone/camera access

Most participants chose the options that the screen does say, ChatGPT may be inaccurate and to not share sensitive info, with 20 participants picking the former and 18 participants picking the latter. However, the options that the screen doesn't explicitly say but they are still policies that ChatGPT does employ were chosen less frequently, with 16 participants saying chats may be used to train models and only 9 participants saying chats may be reviewed by humans. For the people who did choose these options, this could show that these people do have a previous understanding that this is a policy that ChatGPT employs, so that's why they chose this option anyways. However, because less participants chose these options means that this screen doesn't give the full scope of information when it comes to data collection, access, and disclosure, possibly causing a higher number of users to be misinformed about these two policies more so than the policies that Figure 1 *does* list. Moreover, a possible reason that participants understand that chats may be used to train models at a higher rate than the fact that chats may be reviewed by humans, could be because the former is displayed in ChatGPT's privacy

policy [32], which is linked in the figure, however the latter policy is documented in ChatGPT's transparency & content moderation documentation, which is not linked in the figure. This could be a reason why the latter policy is obfuscated in a separate documentation that is not linked, to make it harder for users to access, because this knowledge could impact their decision on letting ChatGPT train on their data or even use the application at all.

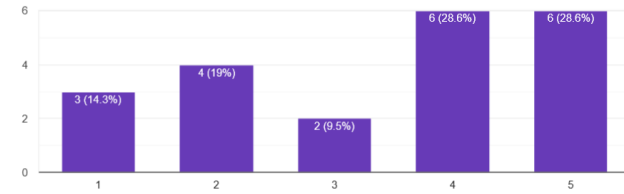
We wanted to be sure whether users understood that their data can be used to train ChatGPT and that their data can be reviewed. So we asked participants if based on Figure 1, will their data be used to train ChatGPT. Most users, 90.5%, understood this policy by answering "Yes". However, when we asked participants who may review the chats between themselves and ChatGPT, 52.4% said they didn't know along with 1 participant who said "no one may review." These results show a disconnect between most users of ChatGPT and who can look at their data, which could've been caused by the obfuscation of these policies within a documentation that's not linked within the welcome page when launching the application.

We also asked participants about how they feel about Figure 1 on a five-level Likert scale. The distribution of these Likert scales and what the survey asked can be seen in Figure 12. Participants mostly gave positive feedback to the Figure 1, however as said before, this figure doesn't give the full information to the users before using ChatGPT, but based on the positive feedback, it can be deduced that any other information put on the welcome screen of ChatGPT could be understood by consumers if displayed in a similar fashion as it is currently displayed. Therefore, this may be a good sign that ChatGPT does have the ability to fully inform users if they present information that is currently only within the documentation in a similar fashion as the figure.

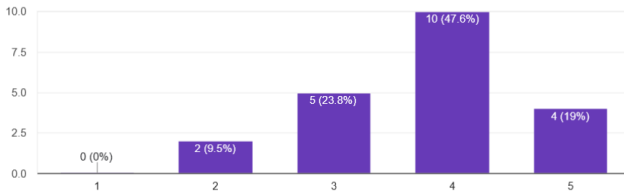
Finally, we asked participants if they know to turn off any permissions that the users implicitly agree to after pressing "Continue." 95.2% of participants said "No." Similar to Apple Intelligence's Siri, this shows a trend where users may agree to start using an application with certain permissions, however if they wanted to later turn off the permissions, they wouldn't know how to, forcing the user to keep using the application in a way the users don't want to or forcing the user to stop using the application all together.

4.4 Does Familiarity Impact User's Behaviors and Expectations?

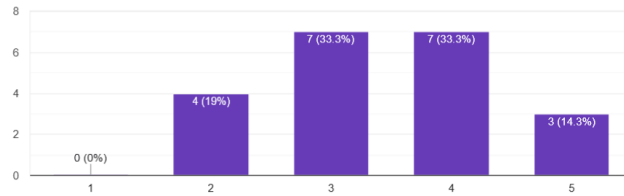
Our survey included participants where some were familiar with the applications that we asked about and where some of the participants weren't familiar with the applications. We wanted to know if there was any relationship between famil-



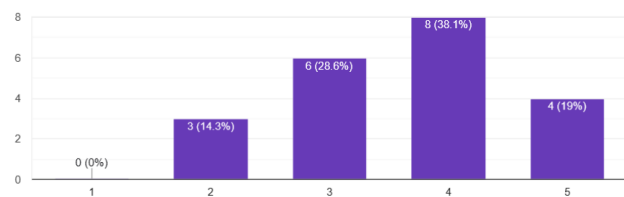
(a) Seeing this screen makes me more cautious about what I type



(b) I am confident I understand what this screen communicates



(c) This screen makes me feel informed before using the app



(d) The warnings are clear and easy to understand

Figure 12: ChatGPT Likert-scale responses

ilarity with the application and how users would/have behaved with the application and their expectations. Due to the small number of responses, we used Fisher's Exact Test to produce p-values between the participants' familiarity and their responses to the questions with a significance value of 0.10, meaning the relationship is significant at $p\text{-val} < 0.10$, because we want these p-values to be more exploratory in its findings, as we find it more important in this study to find *possible* relationships more than the cost of having a false positive. We will define a participant being familiar with the application if they answered 3 or higher on the 5-level Likert scale asking how familiar they are with the application. **There was only 1 participant who said they weren't familiar with ChatGPT, so we decided not to compute any possible significant relationships between familiarity and user's behaviors/expectations because of a lack of data, however we will complete this analysis for Apple Intelligence's Siri.**

4.4.1 Apple Intelligence's Siri

For Apple Intelligence's Siri 10 participants were familiar with the application, while 11 participants weren't. We already know familiarity did not cause participants to fully understand which permissions are defaulted to be on, because 0 users chose all the correct options, nor did it cause users to know how to turn Apple Intelligence's Siri off, because 0 participants chose the correct answer for this as well. We also know it had no effect on whether users think location access is based on functionality, because 100% of users said it needs to be accessed for a request about the weather and 100% said it doesn't need to be accessed for a request like "What is 1+1?"

In Table 1, we compared user's familiarity with the answers they put for the comparison category with the calculated p-val and whether it was determined to be significant or not. Below is how we put people in which bucket for the comparison category if the corresponding question asked in the survey requested a free-response short paragraph answer.

- **Understanding Apple can store user data:** A participant understood if after being presented with Figure 6, their answer mentioned their data being stored
- **Understanding data is reviewed by Apple employee:** A participant understood if their answer *only* mentioned an Apple employee or just mentioned Apple, the entity. They were not considered to understand if they added an extra option on top of that
- **Wanting all permissions to be opt-in:** A participant wanted all permissions to be opt-in if they answered opt-in for all 3 questions asking if a permission should be opt-in or opt-out.

Comparison Category	p-val	Significance
Understanding Apple can store user data	0.5865	Not Significant
Understanding data is reviewed by Apple employees	0.3949	Not Significant
Wanting information to be with the permission itself	0.3108	Not Significant
Wanting all permissions to be opt-in	0.6351	Not Significant

Table 1: Significance Between Familiarity and User Behavior with Comparison Category

Based on this analysis of data, we can conclude that there are no present relationships between a user's familiarity and how they behave or their expectations. So, this can mean that previously mentioned issues are issues that can be applied across the board, regardless if the user is new or old. However, due to the limited size of our study's sample, this is not a conclusive analysis, simply a possibility. This kind of analysis needs to be done in further research to be conclusive.

4.5 Dark Pattern Analysis of Permissions

Based on the survey results, our study has pointed to possible dark patterns within the permissions and information given to users among Apple Intelligence's Siri and ChatGPT. We will be using the definitions defined on the website "Deceptive Patterns" [11].

Apple Intelligence's Siri and ChatGPT had very similar trends when it came to dark patterns within their permissions. The dark patterns that we came across after conducting this survey are:

- **Hard to cancel:** The user finds it easy to sign up or subscribe, but when they want to cancel they find it very hard
- **Obstruction:** The user is faced with barriers or hurdles, making it hard for them to complete their task or access information
- **Preselection:** The user is presented with a default option that has already been selected for them, in order to influence their decision-making
- **Sneaking:** The user is drawn into a transaction on false pretenses, because pertinent information is hidden or delayed from being presented to them
- **Trick wording:** The user is misled into taking an action, due to the presentation of confusing or misleading language

4.5.1 Hard To Cancel

For both Apple Intelligence's Siri and ChatGPT, we asked participants if they knew how to turn Siri off or how to turn off implicitly accepted permissions in ChatGPT after starting to use these applications. This shows a lack of understanding by users on how to turn these applications off due to a lack of presented information. This could be purposefully done by these applications to make it so users have to keep using these application in a way that these applications want them to. This dark pattern isn't as serious as it would have been if these were paid services, but it still makes it difficult for users to control the applications they want to use.

4.5.2 Obstruction

The study showed that some permissions are particularly hard to find or in non-intuitive places, making it difficult for users to control these permissions. In Apple Intelligence's Siri, we saw that the permission to turn Siri was in such a non-intuitive place that no participants chose the correct option when asked how to turn Siri off. Also, some information pertaining to Apple Intelligence's Siri may be hard to find. Our team found multiple articles linked in the settings where information was scattered around. Particularly, there was documentation labeled "About Siri & Privacy..." and "About Improve Siri and Dictation & Privacy...", each with different information. It begs the question about the need for two different types of documentations this kind of separation may be intended to dissuade users from reading all the privacy policies handed to them. People may feel less inclined to read the information if given two different sets of documentations, rather than just 1. And, because this can cause users not to read all the documentation, information can be obfuscated within the documentation, which can mislead users. ChatGPT was seen doing the same thing where important information wasn't even linked within the welcome page, where users probably expect all the information to be given to them, even if in separately linked documents. These applications may be purposefully hiding information within documentation and then hiding the documentation itself within a web of many different documents not linked within the application itself, to mislead and confused users.

4.5.3 Preselection

Both Apple Intelligence's Siri and ChatGPT have a big issue with setting permissions' defaults to be on. And, as seen from the survey, most users do not prefer this to be the case. Most users wanted these permissions to be defaulted to off to give more autonomy to the user and the protect the user's privacy. However, these applications purposefully set these defaults to be on and then don't proceed to ask the users if they would like this permission to be on. This can cause users to use these applications under false pretenses and to allow these

applications access to data that the users don't even know these applications have access to.

4.5.4 Sneaking

This kind of dark permission can be intertwined with preselection and obfuscation. If we define the transaction as the user accepting the terms and conditions of the application to then use it, then based on the survey a lot of participants seemed confused or misunderstood many things that the application did, even if the participant was familiar with the application. For ChatGPT, many participants said they would "Press Continue," even when they didn't fully understand how ChatGPT was using and distributing their data. This could be due to ChatGPT purposefully "sneaking" information away into documentation, so the user will just press continue without understanding the full scope of how their data is accessed and used. Participants also displayed the same type of confusion with Apple Intelligence's Siri, because Siri is doing the same type of "sneaking" as ChatGPT.

4.5.5 Trick wording

We saw this type of dark pattern when presenting participants with screen caps of Apple Intelligence's Siri's policy on being allowed to choose subsets from the users' data to be kept longer than 2 years. Even throughout all of Apple's documentation, we could not find the answers to "How are subsets chosen?" and participants brought up concerns that the wording within the documentation could still allow Apple to define a subset as all of a user's data without any definition of how long, just more than 2 years. This could purposefully be done by Apple to confuse users into allowing Siri access to the user's data under false pretenses.

5 Discussion

Our results show a gap between how Generative AI tools handle privacy and what users expect. We combined our survey data with existing research on dark patterns to identify three main ways that informed consent is compromised in the iOS ecosystem. These are Interface Interference, Semantic Ambiguity, and the gap between Functionality and Surveillance.

5.1 The Failure of "Notice and Consent" and Interface Interference

The "Notice and Consent" model assumes that users will read, understand, and act on privacy policies. Our study shows that this model fails in practice. In our survey, 17 out of 21 participants failed the attention checks inside the Terms and Conditions. This low attention rate matches the work of Felt et al. [13], who showed that users habitually ignore permission dialogs because they see them so often.

However, our results suggest this is not just because users are apathetic. It is also a result of specific design choices. When we showed participants the ChatGPT welcome screen, 94% said they would press "Continue" immediately. The qualitative data links this behavior to the visual design. Specifically, the "Large Continue Button" and "Bold Headings" draw attention away from the "Learn More" links. This confirms the presence of "Interface Interference." Gray et al. [17] define this as a dark pattern where the design visually highlights the easiest path rather than the informed path. By making the "Continue" action dominant and the privacy policy passive, the interface pushes users to skip critical information.

5.2 Semantic Ambiguity: The "Subset" Confusion

An important finding in our study is the confusion around the term "subset" in Apple Intelligence's data retention policy. The policy states that only a "subset" of conversations is reviewed, but it does not define the scope or how that data is chosen.

Our qualitative codebook analysis (Table 5) shows that without clear definitions, users guess incorrectly. Several participants guessed that a "subset" meant "Siri Failures" where the AI failed to answer. Others guessed it meant "mathematical" or "random" samples. The majority simply admitted they did not know. This aligns with observations by Wu et al. [43] regarding "Information Overload." When users see technical but vague documentation, they become overwhelmed and stop paying attention. By using semantic ambiguity, where the terminology is technically accurate but practically vague, the interface appears transparent while hiding the actual privacy risks of human review.

5.3 The Gap Between Functionality and Surveillance

Our results highlight a clear gap between functionality and surveillance in user expectations. When we asked if Siri needs location access for a weather query, 100% of participants correctly said it was necessary for functionality. However, for a logic query like "What is 1+1?", 100% of participants agreed it was *not* necessary.

Despite this clear expectation that data access should depend on the context, the technical reality of iOS permissions is often different. As detailed by Magid's "AppleStorm" research [27], location data is often attached to Siri queries at the system level regardless of the specific prompt. This difference violates the user's trust regarding how their data is used in context. Users believe they are granting "Functional Access" (access only when needed to answer a question). However, the system often interprets this as "Blanket Access" (access for every interaction). This finding supports the concerns raised

by Chitkara et al. [10], who noted that permissions are frequently requested by background processes that are not part of the app’s main features.

5.4 Preference for Opt-In Models

Finally, 19 out of 21 participants preferred an "Opt-in" model for sensitive data training. This contrasts with the current industry standard of "Opt-out" or "Bundled Consent." Participants stated that for sensitive categories like voice recordings and message history, consent should never be assumed. This supports the argument that the current "default-on" settings for model training act as a dark pattern by obstruction. This design forces users to navigate complex settings menus to protect their privacy. Notably, 61.9% of our participants admitted they did not know how to find these menus.

6 Limitations

While this study offers significant qualitative insights, several limitations affect how we interpret our results.

6.1 Sample Size and Demographics

Our study used a small sample size ($N = 21$). This limits the statistical power of our quantitative comparisons. Furthermore, the demographics were unbalanced. 83.3% of participants identified as Asian, and the majority were in the 18–25 age range. Additionally, most participants had a high level of education and a technical background. As noted by Wu et al. [43], technical literacy changes how users read privacy reports. Therefore, our sample likely represents a "best-case scenario" for comprehension. It is likely that the general population faces even greater difficulties in understanding these permissions, though our current data cannot confirm this.

6.2 Ecological Validity and Self-Reporting

We used a survey with static screenshots to simulate user interactions. While this allowed us to control the questions, it does not perfectly match real-time behavior. Self-reported privacy behaviors are often subject to social desirability bias. Although 4 participants claimed they would read the privacy policy, the high failure rate on the attention checks suggests that actual engagement is lower than reported. Future studies would benefit from using instrumented apps to track actual click-through rates and reading times in a live environment.

6.3 Platform Specificity

Our research focused only on the iOS versions of ChatGPT and Apple Intelligence. Android ecosystems use different permission models and UI patterns, as noted in extensive Android-focused literature [8, 13]. Consequently, our specific

findings regarding the "About Siri & Privacy" menu structure may not apply to non-iOS platforms.

7 Conclusion

Adding Generative AI to mobile operating systems creates new privacy risks that current permission frameworks do not fix. Our research shows that the permission prompts in ChatGPT and Apple Intelligence use dark patterns. Specifically, they use Interface Interference and Semantic Ambiguity to lead users into granting more access than they intend.

We identified three key failures in the current design:

- **Visual Manipulation:** Large "Continue" buttons and hidden "Learn More" links stop users from reading privacy policies. This leads to a failure of informed consent.
- **Ambiguous Terminology:** Vague terms like "subset" leave users guessing about how much data humans review. This prevents them from understanding the risk.
- **Contextual Misalignment:** Users expect permissions to be functional and context-aware, such as using location only for weather. However, the system often uses blanket permissions. This creates a breach of trust.

To fix this, we recommend moving from "Opt-out" to "Opt-in" defaults for all model training data. Furthermore, permission prompts must clearly distinguish between **Functional Access** (short-term data use) and **Training Access** (long-term storage). Without these changes, the "dark patterns" in current AI interfaces will continue to weaken user privacy.

8 Conclusion

The integration of Generative AI into mobile operating systems has introduced new privacy risks that current permission frameworks fail to mitigate. Our research demonstrates that the design of permission prompts in ChatGPT and Apple Intelligence employs dark patterns, specifically Interface Interference and Semantic Ambiguity, that manipulate users into granting broader access than they intend.

We identified three key failures in the current design:

- **Visual Manipulation:** Dominant "Continue" buttons and hidden "Learn More" links successfully deter users from engaging with privacy policies, leading to a near-total failure of informed consent.
- **Ambiguous Terminology:** Vague terms like "subset" leave users guessing about the scope of human review, preventing them from accurately assessing risk.
- **Contextual Misalignment:** Users expect permissions to be functional and context-aware (e.g., location only for weather), whereas the system often functions on blanket permissions, creating a breach of trust.

To restore user autonomy, we recommend a shift from "Opt-out" to "Opt-in" defaults for all model training data. Furthermore, permission prompts must evolve to clearly distinguish between **Functional Access** (ephemeral data use) and **Training Access** (persistent storage). Without these changes, the "dark patterns" inherent in current AI interfaces will continue to undermine the ethical foundations of user privacy.

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Primary Code	Sub-Code	Code Operationalization
Asking/Consenting		Used the word "ask" or "consent" to refer to the idea that Apple asks this permission before Siri is able to be used
Functionality		Mentions the idea that Siri needs your location for certain requests to be functional, thus needing to prompt the user to give permission to get their location
Privacy		Mentions the purpose of this permission is for the privacy of the user or to protect the user's privacy rights
Access		Mentions that Siri accesses the location on the user's device
	Separate Access	Mentions that the reason for the permission is for Siri to access something else on the user's device, besides just "location", insinuating that Siri gets the location from somewhere else on the device instead of getting the user's location itself
	Access Data	Mentions that Siri accesses the "phone's data" as a whole to get the location
	Third Party Access	Mentions <i>third party</i>
Tailor Siri		Mentions that location access is used by Siri to tailor Siri's usage to the user based on where the user is located
Doesn't Relate to Siri		Mentions that this permission doesn't relate to Siri or it's functions

Table 2: Codebook for Siri Location Permission Answers

Primary Code	Sub-Code	Code Operationalization
Storage		Mentions that some part or all of the user's data is stored either on the phone or on a separate system
	Apple Sharing	Mentions that the user's data is shared with the entity, Apple, given to Apple, or being held on an Apple-owned system
Improvement		Mentions that the permission is to allow Siri or Apple to use the user's transcripts, voice recording, or other data to improve Siri as a service
	Using	Mentioned use or using in a vague manner without the mention of any storing or specific act of how Siri is improved
	Broad Training or Learning	Mentioned training or learning all of Siri's functionality in general
	Individual Training or Learning	Mentioned training or learning so Siri can be tailored towards the user using it, instead of improving Siri in general
Review		Mentions an entity reviewing a user's data
Access		Mentions Siri or Apple being able to access data besides what's directly given to Siri like audio or transcripts
	Listening	Mentions the permission allows Siri to access other data or parts of the system to always "listen" to the users even without being specifically prompted or without being in use by the user

Table 3: Code book for Siri Training Permission Answers

Primary Code	Sub-Code	Code Operationalization
Technology		Mentions <i>computer</i> or similar device or some sort of technical <i>system</i>
Human		Mentions a human being
	Apple Employee	Mentions that the human is an Apple employee without being specific in what type of employee
	Special Employee	Mentions an Apple employee who is specifically trained to review the user's data or is specifically employed on the Apple Intelligence or Siri team
	Third Party	Mentions <i>third party</i> or people who don't work under Apple
Apple		Mentioned Apple as the company, but does not specify a specific entity under Apple

Table 4: Code book for Who Reviews the User's Data

Primary Code	Sub-Code	Code Operationalization
Dependent On What Is Said		Mentions that the subsets are chosen based on what's being said or how it's being said in audio recordings i.e. tone of voice, language
Textbook Definition		Gives the textbook definition of what a subset is instead of how specifically Siri audio recording and transcripts are picked apart for a subset
Use All/Anything		Mentions that Apple or Siri can or will end up using all of a user's data instead of just a subset
Siri Failure		Mentions that subsets are chosen based on if Siri failed to complete or answer a task and so Siri trains itself on situations it could not understand or wasn't able to fulfill
Documentation Un-said		Mentions that the documentation does not specify or give enough detail for the user to know what a subset is or how a subset is chosen
Improve a Specific Area		Mentions that subsets are chosen based on what Siri currently needs to work on, determined by Siri or Apple itself
	Personal Improvement	Mentions how long the user's data is stored for
Time		Mentions how long the user's data is stored for

Table 5: Code book for What a Subset Is And How It's Chosen

Primary Code	Sub-Code	Code Operationalization
Knowledge		Mentions that whether data should be accessed or permissions should be on or off without the user's knowledge
	Right to Know	Mentions the user's "rights" pertaining to what they should know when Siri is accessing data or what permissions are on or off
Privacy		Mentions user's privacy or right to privacy
Sensitive Data		Mentions the user's sensitive data being a part of the reason why permissions should be opt-in or opt-out

Table 6: Code Book For Why Permissions In Apple Intelligence's Siri should be Opt-In or Opt-Out

A Survey Questions

Here we list the the survey questions and available options.

A.1 Survey Consent

Users were asked to read the consent form and Terms and Conditions, to confirm if they use an iOS device, and to answer the attention check from the Terms and Conditions

A.1.1 Do you consent to participating in this study ?

- I agree

A.1.2 Do you confirm you read the terms and conditions/privacy policy for taking this survey ?

- I confirm

A.1.3 Do you use an IOS mobile device ? (iPhone, iPad)

- Yes
- No

A.1.4 What other generative AI applications do you use ?

Free Response

A.2 Apple Intelligence's Siri Survey

A.2.1 How familiar are you with Siri w/ Apple Intelligence ?

Users were asked to state how familiar they are on a scale of 5 based on the below definitions.

- 1: I have never downloaded Apple Intelligence or I have but I never used Siri after downloading
- 5: I downloaded Apple Intelligence and use Siri often

A.2.2 What permissions do you think or that you are aware of that Siri has that are automatically turned ON after downloading Apple Intelligence ?

Multi-select question with an "other" option, where participants are allowed to answer with their own written response.

- Siri may train itself based on your data
- Transcripts and Audio Transcripts between you and Siri are allowed to be stored on Apple Servers
- Location Access
- Siri may listen passively at all times in case you may need them

- All transcripts and audio transcripts must stay locally on your device
- Siri may look at what apps you're using to learn more about you
- Transcripts of all search requests you ask Siri(i.e. asking Siri to look something up on Safari)

A.2.3 Are there any permissions you specifically turned on or off after downloading Apple Intelligence that pertain to Siri ? Explain why you turned it off or on. You may say "None" if not applicable to you.

Free Response

A.2.4 Do you know how to turn off Apple Intelligence's Siri after downloading it ?

Multi-select question with an "other" option, where participants are allowed to answer with their own written response.

- Turn it off in Privacy Security Settings
- Turn it off in Siri Settings
- Turn it off in Content Privacy Restrictions settings in Screen Time settings
- I don't know, but I would search online

A.2.5 This is a permission that relates to Apple Intelligence's Siri. What do you think it means to have it on ?

Free Response

A.2.6 Do you think Siri needs to access your location for the following question? "What is the weather today ?"

- Yes
- No

A.2.7 Do you think Siri needs to access your location for the following question? "What is 1+1 ?"

- Yes
- No

A.2.8 This is a permission that relates to Apple Intelligence's Siri. What do you think it means to have it on ?

Free Response

A.2.9 This is a screenshot from Apple Intelligence's Siri, Dictation, Privacy page. Who do you think your information is reviewed by ?

Free Response

A.2.10 This screenshot pertains to Apple's being able to store your transcripts and audio recordings. From this description, can you define what a subset could be and how a subset is chosen ?

Free Response

A.2.11 Pertaining to the images in the 2 above questions, do you wish for the information in Apple's documentation to be showed with the permission itself(i.e. in the short caption under it)? Or, is it within Apple's right to put it in a separate document for users to choose whether to read it or not ?

- I wish for the information in Apple's documentation to be showed with the permission itself
- It is within Apple's right to put it in a separate document for users, thus it is the user's responsibility

A.2.12 For any possible permissions that may pertain to Apple Intelligence's Siri, do you think they should be Opt-in or Opt-out? And, why ?

Users were given this description:

Opt-in = They should automatically be turned off and you can choose whether to turn it on

Opt-out = They should be automatically turned on and you can choose whether to turn it off

Free Response

A.2.13 What about permissions relating to Apple being able to send your data outside of your device? If it depends, fill out what it depends on in the "Other" option.

Users were given an "other" option, where they could answer with their own response

- Opt-in
- Opt-out

A.2.14 What about permissions for Siri to have access to information that Siri doesn't need to function but may make Siri "work better"? If it depends, fill out what it depends on in the "Other" option.

Users were given an "other" option, where they could answer with their own response

- Opt-in
- Opt-out

A.3 ChatGPT Survey

A.3.1 Have you downloaded and used the ChatGPT app on your IOS mobile device ?

- Yes
- No

A.3.2 How familiar are you with ChatGPT in general ?

Users were asked to state how familiar they are on a scale of 5 based on the below definitions.

- 1: I have never used ChatGPT
- 5: I use ChatGPT very often

A.3.3 What's better — ChatGPT or other AI applications?

Free Response

A.3.4 What would you do right now?

Users were given a screenshot of ChatGPT's mobile application welcome screen

Multi-select question

- Press Continue
- Close the app
- Click "Learn more"
- Click "Terms"
- Click "Privacy Policy"

A.3.5 What is this screen mainly telling you?

Users were given a screenshot of ChatGPT's mobile application welcome screen

Multi-select question with an "other" option, where participants are allowed to answer with their own written response

- ChatGPT may be inaccurate
- Don't share sensitive info
- Chats may be reviewed by humans
- Chats may be used to train models
- It needs microphone/camera access
- None of the above / Not Sure

A.3.6 According to this screen, will your chats be used to train models?

Users were given a screenshot of ChatGPT's mobile application welcome screen

- Yes
- No

A.3.7 According to this screen, who may review the chats between yourself and ChatGPT?

Users were given a screenshot of ChatGPT's mobile application welcome screen

- Only automated systems
- Human reviewers may
- No one reviews
- I am not sure

A.3.8 Where would you expect to learn more details about data use?

Users were given a screenshot of ChatGPT's mobile application welcome screen

- The "Learn more" link
- Settings after onboarding
- External help center
- Nowhere obvious
- Not sure

A.3.9 I am confident I understand what this screen communicates.

Users were asked to state how much they understand on a scale of 5 based on the below definitions.

- 1: Not at all
- 5: Very

A.3.10 The warnings are clear and easy to understand.

Users were asked to state how much they understand on a scale of 5 based on the below definitions.

- 1: Not at all
- 5: Very

A.3.11 This screen makes me feel informed before using the app.

Users were asked to state how much they feel informed on a scale of 5 based on the below definitions.

- 1: Not at all
- 5: Very

A.3.12 Seeing this screen makes me more cautious about what I type.

Users were asked to state how much they feel they are more cautious on a scale of 5 based on the below definitions.

- 1: Not at all
- 5: Very

A.3.13 What parts of the design influence your decision making?

Multi-select question with an "other" option, where participants are allowed to answer with their own written response

- Large "Continue" button
- "Learn more" link placement
- Iconography (flag/lock)
- Bold headings
- Dark theme
- None

A.3.14 Do you know how to turn off this permission after you click "Continue"

- Yes
- No

A.3.15 If you answered "Yes" in last question, please briefly tell the process and why you want to turn off it.

Free Response

A.4 Demographics and Follow-up

A.4.1 How do you spend your free time?

Free Response

A.4.2 What parts of the design influence your decision making?

Multi-select question

- Any variation of the iPhone 17
- iPhone Air
- Any variation of the iPhone 16
- iPhone 15 Pro or iPhone 15 Pro Max
- iPad Pro
- iPad Air
- iPad mini
- Older/Different Model

A.4.3 Do you have IOS 26 on your Apple mobile device ?

- Yes
- No

A.4.4 What is your gender identification ?

- Woman
- Man
- Non-Binary
- Other
- I do not wish to provide this information

A.4.5 What is your age ?

- 18-25
- 26-35
- 36-50
- 50+
- I do not wish to provide this information

A.4.6 What is your age ?

Multi-select

- White
- Asian
- Black or African American
- Hispanic or Latino
- Native American or Alaska Native
- Native Hawaiian or Other Pacific Islander
- I do not wish to provide this information

A.4.7 Are you a part of the LGBTQIA+ community ?

- Yes
- No
- I do not wish to provide this information

A.4.8 How technical would you say your profession or focus of study is ? You may skip this question if you do not wish to provide this information.

Users were asked to state how technical it is on a scale of 5 based on the below definitions

- 1: Not technical at all
- 5: Extremely technical

A.4.9 How much is your yearly income ?

Users were given this description:

If you are being supported by someone else or by other means and have your own income, please put your best guess of your support's income, plus your own.

- 0-15,999\$
- 16,001\$ - 29,999\$
- 30,001\$ - 49,999\$
- 50,001\$ - 74,999\$
- 75,000\$ - 99,999\$
- 100,000\$ - 199,999\$
- 200,000\$+
- I do not have an income
- I do not wish to provide this information

A.4.10 What is your education level(including in progress) ?

- High School
- Associate's Degree
- Bachelor's Degree
- Master's Degree
- P.H.D.
- None
- I do not wish to provide this information

B Debrief Survey

B.0.1 This is a screenshot taken from our Terms and Conditions, when you saw this question, did you answer with what the documentation said to, or close to it?

Users were given a screenshot of the first attention check from the Terms and Conditions

- Yes
- No

B.0.2 If no, why ?

Users were given an "other" option, where participants are allowed to answer with their own written response.

- I didn't read the Terms and Conditions
- I skimmed over it so I didn't catch any details
- I forgot it said that, but I read the Terms and Conditions

B.0.3 This is a screenshot taken from our Terms and Conditions, when you saw this question, did you answer with what the documentation said to, or close to it?

Users were given a screenshot of the second attention check from the Terms and Conditions

- Yes
- No

B.0.4 If no, why ?

Users were given an "other" option, where participants are allowed to answer with their own written response.

- I didn't read the Terms and Conditions
- I skimmed over it so I didn't catch any details
- I forgot it said that, but I read the Terms and Conditions

B.0.5 This is a screenshot taken from our Terms and Conditions, when you saw this question, did you answer with what the documentation said to, or close to it?

Users were given a screenshot of the third attention check from the Terms and Conditions

- Yes
- No

B.0.6 If no, why ?

Users were given an "other" option, where participants are allowed to answer with their own written response.

- I didn't read the Terms and Conditions
- I skimmed over it so I didn't catch any details
- I forgot it said that, but I read the Terms and Conditions

B.0.7 Please read the debrief document. Do you still consent to the researchers keeping and using your response to this survey ?

- Yes
- No